

Python Mastery: File Handling

Opening, Reading, Writing, and Managing Files

1. Introduction to File Handling

File handling is an important part of any web application. Python has several functions for creating, reading, updating, and deleting files.

The open() Function

The key function for working with files in Python is the `open()` function. It takes two parameters: **filename** and **mode**.

Mode	Description
"r" - Read	Default value. Opens a file for reading; error if it doesn't exist.
"a" - Append	Opens a file for appending; creates the file if it doesn't exist.
"w" - Write	Opens a file for writing; creates the file if it doesn't exist.
"x" - Create	Creates the specified file; returns an error if it exists.

2. Reading Files

To open a file for reading, use the "r" mode. You can read the whole content or just specific parts.

```
f = open("demofile.txt", "r")
print(f.read(5)) # Read first 5 characters
print(f.readline()) # Read one line
f.close()
```

3. Writing and Creating Files

To write to an existing file, you must add a parameter to the `open()` function: "a" for append or "w" for overwrite.

```
f = open("demofile.txt", "a")
f.write("Now the file has more content!")
f.close()

# Overwrite the content:
f = open("demofile.txt", "w")
f.write("Woops! I have deleted the content!")
f.close()
```

4. Deleting Files

To delete a file, you must import the **OS module** and use its `os.remove()` function.

```
import os
if os.path.exists("demofile.txt"):
    os.remove("demofile.txt")
else:
    print("The file does not exist")
```

5. Best Practice: The 'with' Statement

It is good practice to use the `with` keyword when dealing with file objects. The advantage is that the file is properly closed after its suite finishes.

```
with open("test.txt", "w") as f:
    f.write("Hello World")
# File is automatically closed here
```